Product Optimization and Revenue Analysis for Afficionado Coffee Roasters

Abstract - This project focuses on analyzing product-level sales performance for afficionado Coffee Roasters using transaction data. The objective is to identify high-performing products, understand category-level revenue distribution, and evaluate the relationship between product popularity and revenue generation. Exploratory Data Analysis (EDA) techniques and interactive visualizations were used to uncover business insights. The analysis revealed that a small number of products contribute a significant share of total revenue and that revenue generation is concentrated within a few product categories. The finding supports data-driven decision making for product optimization, inventory management, and revenue growth.

Keywords:

Data analytics, Product Optimization, Revenue Analysis, Coffee Retail, Streamlit, Exploratory Data Analysis

Introduction

Understanding product performance is essential for maximizing profitability in the coffee retail industry. Businesses must identify which products generate the highest revenue, attract the most customers, and contribute significantly to overall sales. Data-driven product analysis helps organizations optimize inventory planning, improve menu offerings and focus on products that deliver the greatest business value.

This project analyzes transaction data from Afficionado Coffee roasters to evaluate product performance across categories and product types. The study aims to identify revenue-generating products, analyze category contributions, and provide actionable recommendations for business optimization.

Objective

The primary objectives of this project are to analyze the product portfolio of Afficionado Coffee Roasters by identifying the top-selling and least selling products, quantifying the revenue contribution of each product and category, and measuring revenue concentration across the menu. This will provide a clear understanding of product performance and sales distribution.

Secondary objective is to support menu simplification and optimization based on sales and revenue data, Identify high-impact "hero" products that drive significant revenue. Highlighting low-performing products for review, potential redesign, or removal.

Data Collection

The data used for this project was collected from Afficionado Coffee Roasters sales records over a specified period. It includes detailed transaction-transaction level information such as product names quantities sold sale dates and revenue generated for sale additional data was source from inventory and renew details to categorize products appropriately the data was extracted in a structured format portable for analysis in Python, ensuring accuracy and completeness for reliable insights.

Methodology

This project was conducted using Python based data analysis and visualization techniques.

The dataset was initially loaded and examined for data quality issues, including missing values and consistencies.

Listing 1: Data Ingestion and Cleaning

```
# Data Loading and Cleaning
df=pd.read_csv("Afficionado Coffee
Roasters.csv")

df.drop_duplicates(inplace=True)
df.dropna(inplace=True)
```

Relevant variables were selected to focus on product performance analysis. A new revenue column was created by multiplying the transaction quantity by the unit price. This metric served as the basis for evaluating product and category performance throughout the analysis.

Listing 2: Sample Python code for revenue calculation

```
# Calculate revenue per transaction
```

```
df['Revenue']=df['transaction_qty']*[df]['unit_price']
```

Category level analysis showed that revenue generation was concentrated within a within a limited number of product categories. This indicates that certain categories play a more significant role in overall profitability than others.

Listing 3: Category wise Revenue analysis

```
# Calculating Revenue generation by Category
rev_pro_category=df.groupby("product_category")['revenue'].sum()
```

Exploratory Data Analysis (EDA) was performed using Pandas, Matplotlib, Seaborn, and Plotly.

```
# Visualizing Revenue Contribution by Product
import matplotlib.pyplot as plt
rev_pro.sort_values(ascending=False).head(10).plot(kind='bar',title="Revenue Contribution by Product")
plt.xlabel("Product")
plt.ylabel("Revenue")
plt.show()
```

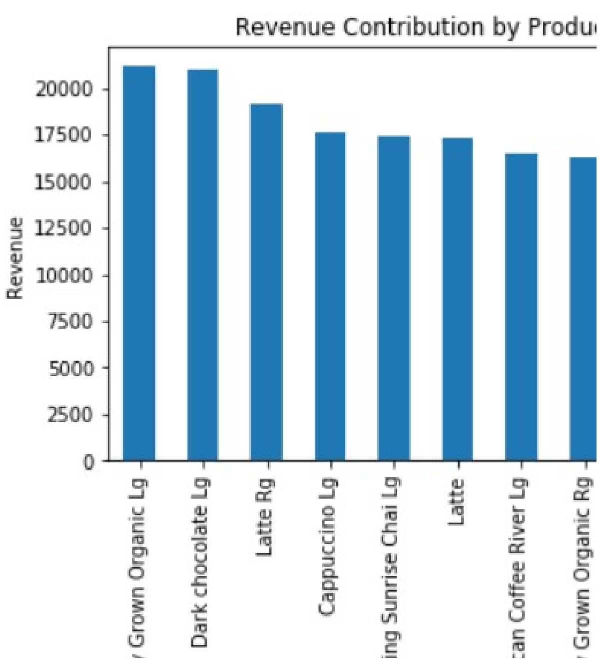

FIGURE 1: REVENUE GENERATION BY PRODUCT

Product rankings were generated based on sales volume and revenue contribution.

```
# Product Ranking
rank_sort=unit_pro.sort_values(ascending=False).head(10).plot(kind='bar')
plt.xlabel("Products")
plt.ylabel("Sales")
plt.title("Top Products by Sales Volume")
```

Scatter plots were employed to explore the relationship between revenue generation and product popularity. Additionally product drill-down analysis facilitated detailed evaluation of individual product performance.

```
# Comparison b/w volume rank and revenue rank
import matplotlib.pyplot as plt

x=unit_pro
y=rev_pro

plt.scatter(x,y)
plt.axhline(y.mean(), linestyle='--')

plt.xlabel("Quantity Sold")
plt.ylabel("Revenue")

plt.title("Popularity vs Revenue")
plt.show()
```

An interactive Streamlit dashboard was developed, offering real-time visualizations, product and category filters, location-based analysis, and performance monitoring. This dashboard enables users to explore key business metrics and make informed, data-driven decisions.

```
# Creating  interactive product optimization dashboard
Import streamlit as st
st.title("Product Optimization Dashboard")
st.dataframe(df.head)
```

Analysis and Results

A. Revenue Contribution Analysis

The dataset was analyzed to identify product level performance patterns and revenue contribution across categories.

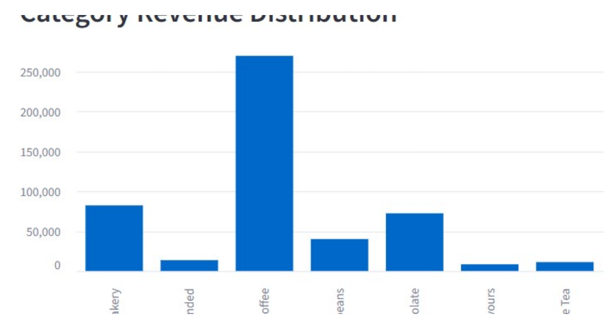

FIGURE 1. REVENUE CONTRIBUTION BY PRODUCT CATEGORY

Coffee category generated the highest revenue, followed by bakery products. Packed products contributed comparatively less revenue. This indicates that customers showed stronger preference toward beverage-based offerings.

B .Product Ranking Analysis

Product ranking analysis revealed that a small number of products contributed a significant share of overall revenue. This product consistently demonstrated strong sales volume and customer demand, making them key drivers of business performance

Top 10 products by revenue

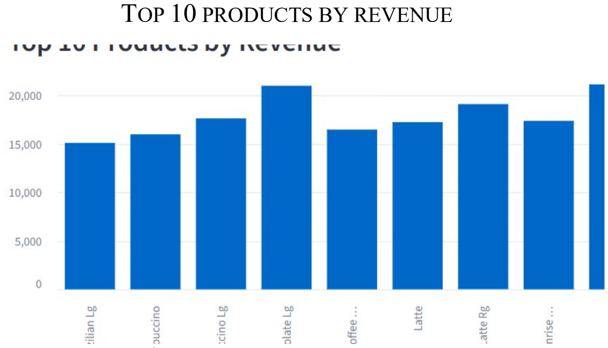

FIGURE 2. TOP REVENUE GENERATING PRODUCTS

The popularity versus revenue analysis demonstrated a generally positive relationship between product demand and revenue generation.

Analysis of popularity and revenue

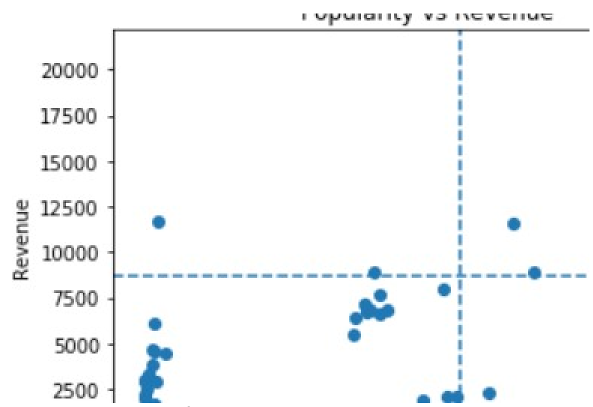

FIGURE 3: POPULARITY VS REVENUE

Products with higher sales volume often generated greater revenue, although some premium products achieved substantial revenue despite lower transaction volumes.

Revenue share per category analysis highlights the contribution of each product category to the overall revenue generated by the business.

Revenue share per category

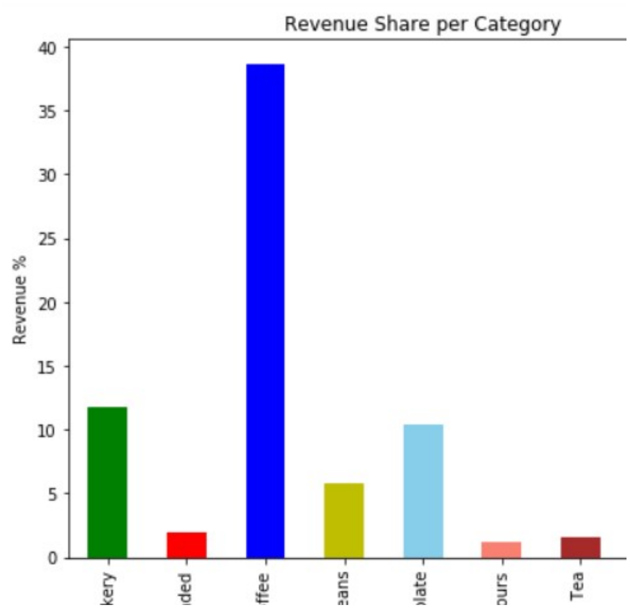

FIGURE 4: REPRESENTING REVENUE SHARE BY CATEGORY

The results indicate that a few categories account for a significant portion of total sales, reflecting strong customer demand and purchasing preferences. Categories with higher revenue contributions represent key business drivers and may benefit from focused marketing and inventory management strategies.

The Pareto analysis was conducted to identify the products contributing the most revenue.

PARETO ANALYSIS (80/20 RULE)

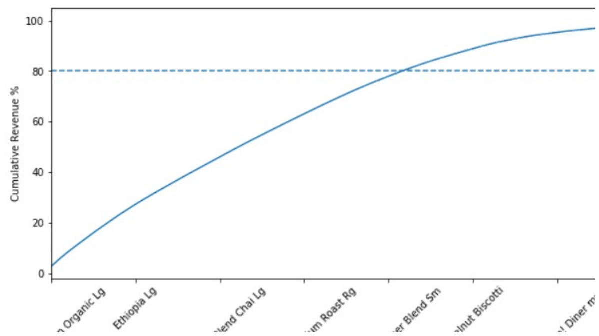

FIGURE 5: REVENUE CONCENTRATION ANALYSIS

The results showed that a small percentage of products generated a large share of total revenue, supporting the Pareto principle (80/20 rule). This finding suggests that focusing on high-performing products can significantly impact overall business performance, inventory planning, and revenue growth.

Visualization techniques including bar charts, category distribution charts, scatter plots and product performance tables were used to identify trends and compare product performance across multiple dimensions.

System Implementation using Streamlit

The Streamlit dashboard provided an interactive platform for exploring product rankings, category performance, store-level insights, and revenue metrics through dynamic filters and visualizations.

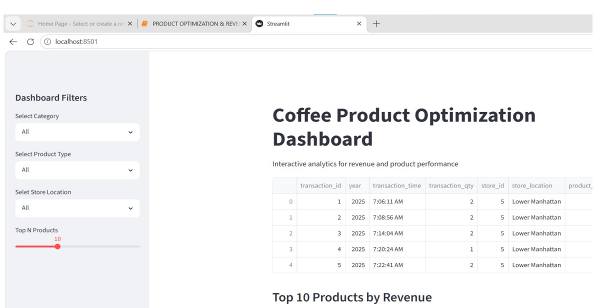

FIGURE 4. INTERACTIVE PRODUCT OPTIMIZATION DASHBOARD

DISCUSSION

The analysis indicates that revenue generation is highly concentrated among a limited number of products and categories. This suggests that certain products play a critical role in overall business performance and should receive greater attention in inventory planning and promotional strategies.

The positive relationship between product popularity and revenue demonstrates that customer demand significantly influences sales performance stop. However, the presence of high revenue products with relatively lower sales volume indicates opportunities for premium product positioning and targeted marketing.

The findings highlight the importance of continuously monitoring product performance to identify imaging trends emerging trends and changing customer preferences. Businesses can use these insights to optimize product offerings, improve resource allocation and maximize profitability.

The results successfully address the project objectives by identifying key revenue drivers, evaluating category performance and providing a foundation for data-driven decision making.

RECOMMENDATIONS

Based on the analysis, the following recommendations are proposed:

- Focus inventory planning on high performing products to ensure consistent product availability.
- Increase promotional efforts for products that generate high revenue and strong customer demand.
- Explore opportunities to improve the performance of lower selling products through targeted marketing and product bundling strategies.
- Reduce dependence on a small number of products by diversifying the product portfolio and introducing new offerings.
- Continuously monitor product and category performance using data-driven dashboards to support informed business decisions.
- Utilize customer purchasing trains to optimize pricing, inventory allocation and menu planning.

CONCLUSION

The project successfully analyzed product level sales performance for afficionado coffee roasters using transaction data and data visualization techniques. The study identified key revenue generating products, evaluated category-wise revenue contribution, and examined the relationship between product popularity and revenue generation let's stop.

The findings revealed that a small number of products contribute significantly to overall business revenue and the revenue generation is concentrated within a limited number of product categories. These insights provide valuable support for inventory management, product optimization, and strategic decision making.

The Streamlit dashboard for the enhanced analysis by providing an interactive platform for monitoring product performance and exploring business insights. Overall, the project demonstrates the value of data analytics in improving operational efficiency and supporting business growth.

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